

Introduction to Physical Modelling for Sound Synthesis

Differentiable Physical Modelling and Machine Learning for Sound Synthesis

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2026-05-07

Week 1 – Introduction to Physical Modelling

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github.com/rodrigodzf/physmod-ml-course



Overview

- Practical information: logistics, schedule, resources
- Physics-based vs. data-driven approaches
- Two views of physics-based modelling, and the spectral baseline
- Continuous physics: PDEs, boundary and initial conditions
- Free vs. forced systems, damping
- Case study: the vibrating string
- The real-time constraint
- State-space form and nonlinearities
- Numerical simulation: time-stepping and stability

Practical Information

Logistics

- **Room:** H 3001
- **Time:** 14–16, weekly from 7 May 2026
- **Credits:** 3 ECTS
- **Tools:** Python, JAX, PyTorch

Important dates:

- **Course application deadline:** 18.05.2026
- **Project proposal deadline:** 07.06.2026
- **Project pitch:** 25.06.2026
- **Final presentation:** 09.07.2026
- **Code + paper submission:** 30.09.2026

Course Roadmap

Week	Topic
1	Introduction to Physical Modelling (today)
2	Modal and Finite-Difference Methods
3	Nonlinear Systems and Coupled Models
4	Differentiable Programming for Audio
5	Neural ODEs
6	Physics-Informed Neural Networks
7	Neural Operators (FNO)
8	Deep State-Space Models and Koopman Theory
9	Differentiable Physical Modelling for Parameter Estimation
10	Project Presentations

Resources

Physics and numerical simulation

- Bilbao, *Numerical Sound Synthesis: Finite Difference Schemes and Simulation in Musical Acoustics* (Wiley, 2009) – main reference for FDTD and energy methods.
- Fletcher and Rossing, *The Physics of Musical Instruments* (Springer, 2nd ed., 1998) – physical foundations.
- Trautmann and Rabenstein, *Digital Sound Synthesis by Physical Modeling Using the Functional Transformation Method* (Springer, 2003) – FTM and transform-based synthesis.
- Smith, *Physical Audio Signal Processing* – ccrma.stanford.edu/~jos/pasp/

Combining physics and neural networks

Physics-based Audio vs. Data-driven Approaches

Two Paradigms

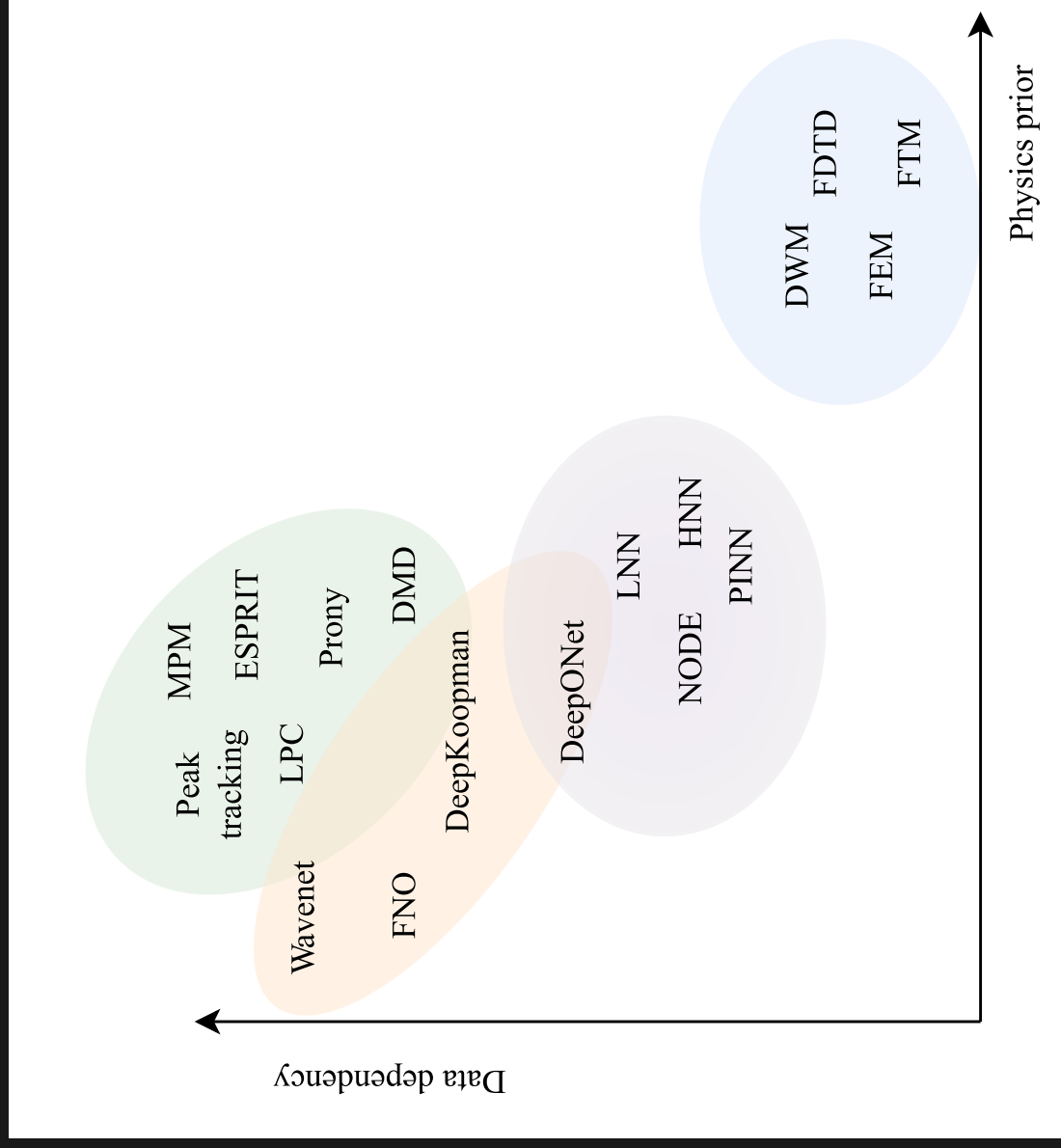
Physics-based

- No audio samples used
- Sound generated from mathematical description
- Compact and lightweight, often just kilobytes
- Recursive systems at audio rate (IIR)
- Deterministic

Data-driven

- Learns from recorded audio
- Requires large datasets
- Models are large (MB–GB)
- Feed-forward or autoregressive
- Requires training and generalisation

Two Paradigms



Week 1 – Introduction to Physical Modelling

What is a Model?

A model is a simplified representation of a system that lets us predict its behaviour.

Physics-based model Data-driven model

Derived from first principles (governing equations).

$$\dot{\mathbf{h}} = f(\mathbf{h}) + e(t)$$

Learned from observations (input-output pairs).

$$\mathbf{y} = f_{\theta}(\mathbf{h}, e(t))$$

Both are approximations, they differ in where the knowledge comes from: **physical laws** vs. **data**

Note: physics-based models can be *equation-based* (simulate the ODE/PDE numerically) or *solution-based* (evaluate a closed-form solution, e.g. $w(t) = A \cos(\omega t + \varphi)$). In practice, most systems of interest have no closed-form solution, so this course

Physics-based Audio: The goal

direct generation of high-quality, acoustic sound through numerical simulation.

System families:

- 1D – strings, bars, acoustic tubes
- 2D – membranes, plates
- 3D – enclosures, solid bodies

Applications:

- Synthesis (brass, percussion, strings)
- Effects (reverb, room simulation)

Approaches to Physics-based Modelling

Two Views of Physics-based Modelling

Spectral Models

Assume the signal can be approximated as a sum of damped sinusoids:

$$y(t) = \sum_{k=1}^K A_k e^{-\alpha_k t} \cos(\omega_k t + \varphi_k)$$

Synthesis side: additive synthesis, subtractive synthesis, resonant filter banks.

Analysis side: spectral peak tracking, Prony's method, LPC, ESPRIT, DMD.

Cheap, interpretable, real-time friendly, and a workhorse of audio DSP since the 1980s.

Why these aren't physical models

- Parameters (ω_k , α_k , A_k) are fitted to a signal, not derived from governing equations.
- No notion of geometry, material, boundary conditions, or excitation mechanism.
- Typically **single-point** in space, no spatial field $w(\mathbf{x}, t)$.
- Linear and time-invariant by construction, struggles with strong nonlinearity (tension modulation, collisions) and time-varying behaviour.
- Describes **what the sound looks like**, not **what produced it**.

They sit alongside physical models as a useful baseline, and reappear later as the linear backbone of modal synthesis and Koopman-style learned models.

From Continuous Physics to Discrete Computation

Governing Equations

State is a field $w(\mathbf{x}, t)$ over a spatial domain Ω (string, membrane, air column), governed by a PDE.

1D wave equation:

$$\frac{\partial^2 w}{\partial t^2} = \hat{T} \frac{\partial^2 w}{\partial x^2}$$

w : displacement. $\hat{T}$: wave speed.

Higher dimensions: replace $\partial^2 / \partial x^2$ with the Laplacian Δ .

Drop spatial dependency $\Rightarrow$ ODE (lumped oscillator).

Boundary Conditions

A PDE alone is not enough. We must specify what happens **at the edges** of the domain.

For a string of length L , common boundary conditions:

Type	Condition	Physical meaning
Fixed (Dirichlet)	$w(0, t) = w(L, t) = 0$	Clamped ends (e.g. guitar string at nut and bridge)
Free (Neumann)	$\frac{\partial w}{\partial x}(0, t) = \frac{\partial w}{\partial x}(L, t) = 0$	No transverse force at the boundary

Boundary Conditions



The choice of boundary conditions determines the **mode shapes** and **eigenfrequencies** of the system. When we discretise, BCs affect the structure of the resulting matrices (e.g. the corner entries of a discrete Laplacian). More on this in week 2.

Initial Conditions

We also need to specify the state of the system **at time** $t = 0$.

For the wave equation (second order in time), we need two initial conditions:

$$w(x, 0) = w_0(x) \quad (\text{initial displacement})$$

$$\frac{\partial w}{\partial t}(x, 0) = v_0(x) \quad (\text{initial velocity})$$

For sound synthesis, we typically start **at rest**:

$$w_0(x) = 0, \quad v_0(x) = 0$$

and drive the system via an external force.

Free vs. Forced Systems

Why Synthesis Requires the Forced Form

A free system: $\dot{\mathbf{h}} = f(\mathbf{h})$, with initial conditions $\mathbf{h}(0) = \mathbf{h}_0$.

But in sound synthesis, the system starts **at rest**: $\mathbf{h}(0) = \mathbf{0}$.

Sound is initiated by an **external driving function**:

$$\dot{\mathbf{h}} = f(\mathbf{h}) + g(t)$$

This represents the physical interaction with the object (pluck, strike, bow, breath).

Damping and Loss

Real systems are never truly lossless. We add damping terms to the wave equation:

$$\frac{\partial^2 w}{\partial t^2} = \hat{T}^2 \frac{\partial^2 w}{\partial x^2} + \delta(x - x_i) f(t)$$

Undamped – oscillations persist forever.

Damping and Loss

Real systems are never truly lossless. We add damping terms to the wave equation:

$$\frac{\partial^2 w}{\partial t^2} = \hat{T}^2 \frac{\partial^2 w}{\partial x^2} - \hat{d}_1 \frac{\partial w}{\partial t} + \hat{d}_3 \frac{\partial^3 w}{\partial t \partial x^2} + \delta(x - x_i) f(t)$$

- $\hat{d}_1$: frequency-independent damping (all modes decay equally)
- $\hat{d}_3$: frequency-dependent damping (high frequencies decay faster – realistic)

Even small damping profoundly affects decay and tonal quality. Musical instruments are “high-Q” systems. Low loss, but perceptually critical.

Case Study: The Vibrating String

The Wave Equation

The ideal vibrating string – second-order linear wave equation:

$$\frac{\partial^2 w}{\partial t^2} = \frac{T}{\rho \pi r^2} \frac{\partial^2 w}{\partial x^2}, \quad x \in [0, L]$$

After non-dimensionalisation (reducing 4 parameters to 1):

$$\frac{\partial^2 w}{\partial t^2} = \hat{T}^2 \frac{\partial^2 w}{\partial x^2}, \quad x \in [0, 1]$$

Non-dimensionalisation is essential for efficient analysis and crucial for ML-based parameter fitting.

Excitation: Pluck vs. Strike

Pluck

- Initial condition on displacement
- Often a triangle shape
- Parameters: position, amplitude

Strike

- External driving function $f(t)$
- Applied at location x_i
- E.g. raised cosine pulse

$$\frac{\partial^2 w}{\partial t^2} = \hat{T}^2 \frac{\partial^2 w}{\partial x^2} + \delta(x - x_i) f(t)$$

Towards a Realistic String Model

Adding stiffness ($\hat{D}$), damping ($\hat{d}_1, \hat{d}_3$), and geometric nonlinearity (K):

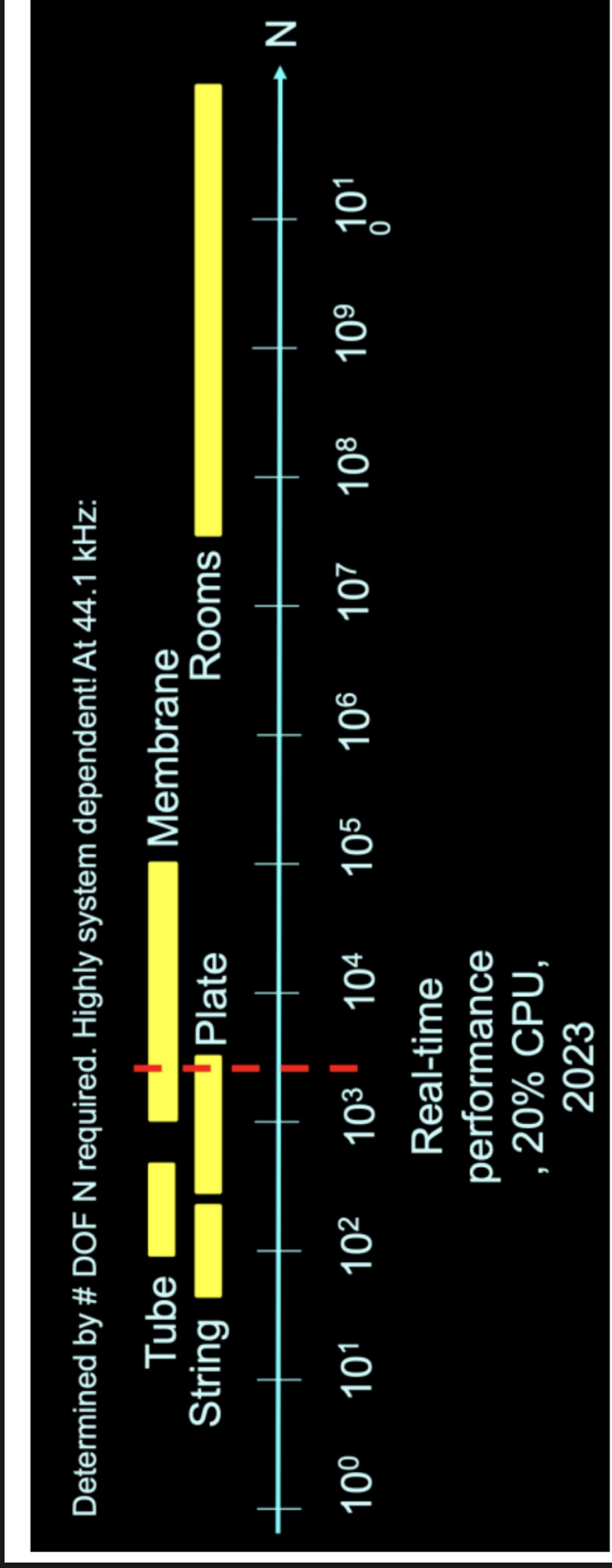
$$\frac{\partial^2 w}{\partial t^2} = \hat{T}^2 \frac{\partial^2 w}{\partial x^2} - \hat{D} \frac{\partial^4 w}{\partial x^4} - \hat{d}_1 \frac{\partial w}{\partial t} + \hat{d}_3 \frac{\partial^3 w}{\partial t \partial x^2} + K^2 \left(\frac{\partial w}{\partial x} \right)^2 \frac{\partial^2 w}{\partial x^2}$$

Term	Effect
$\hat{D}^2 \partial^4 w / \partial x^4$	Stiffness → inharmonicity (dispersion)
$\hat{d}_1 \partial w / \partial t$	Frequency-independent damping
$\hat{d}_3 \partial^3 w / \partial t \partial x^2$	Frequency-dependent damping (realistic)
$K^2 (\partial w / \partial x)^2 \partial^2 w / \partial x^2$	Tension modulation → pitch glide, brightness

The Real-time Constraint

Fixed Step-size Simulation at Audio Rate

Human hearing extends to ~ 20 kHz, so by Nyquist $T < 1/(2f_c) \approx 25 \mu\text{s}$. One sample per time step, at audio rate (44.1 / 48 kHz).



Source: Bilbao, *Physics-based Audio: Sound Synthesis and Virtual Acoustics*, 2023.

State-space Representation

From 2nd-order PDE to State-space

Start from the (forced, damped) 1D wave equation:

$$\frac{\partial^2 w}{\partial t^2} = \hat{T}^2 \frac{\partial^2 w}{\partial x^2} - 2\hat{d}_1 \frac{\partial w}{\partial t} + b(x) f(t)$$

Spatial discretisation – replace $\partial^2 / \partial x^2$ by a discrete Laplacian $\mathbf{D}_{xx}$ on a grid. Let $\mathbf{w}(t) \in \mathbb{R}^N$ stack the values $w(x_i, t)$:

$$\ddot{\mathbf{w}} = -\mathbf{K}\mathbf{w} - 2\hat{d}_1 \dot{\mathbf{w}} + \mathbf{b} f(t), \quad \mathbf{K} = -\hat{T}^2 \mathbf{D}_{xx}$$

A 2nd-order system of N coupled ODEs.

Lift to first order – stack $\mathbf{w}$ and $\dot{\mathbf{w}}$ into a state vector:

$$\frac{d}{dt} \begin{bmatrix} \mathbf{w} \\ \dot{\mathbf{w}} \end{bmatrix} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{K} & -2\hat{d}_1 \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{w} \\ \dot{\mathbf{w}} \end{bmatrix} + \begin{bmatrix} \mathbf{0} \\ \mathbf{b} \end{bmatrix} f(t)$$

The first block row is the trivial $\dot{\mathbf{w}} = \dot{\mathbf{w}}$; the second is the original PDE. Nonlinearities, when present, sit in the second block row.

The Nonlinear State-space Form

The state $\mathbf{h}$ stacks displacement and velocity:

$$\mathbf{h} = \begin{bmatrix} \mathbf{w} \\ \dot{\mathbf{w}} \end{bmatrix} \in \mathbb{R}^{2N}$$

$$\begin{aligned} \dot{\mathbf{h}} &= \mathbf{A}\mathbf{h} + f_{\text{NL}}(\mathbf{h}) + \mathbf{b} f(t) \\ \mathbf{y} &= \mathbf{C}\mathbf{h} \end{aligned}$$

Term	Role
$\mathbf{A}\mathbf{h}$	Linear dynamics
$f_{\text{NL}}(\mathbf{h})$	Nonlinear dependencies on state
$\mathbf{b} f(t)$	External driving signal
$\mathbf{C}\mathbf{h}$	Output mapping (linear sampling of the state)

Types of Nonlinearities in Physical Models

Physical phenomenon	Mathematical form
Tension modulation (strings, membranes)	Cubic polynomial
Collision (mallet-drum, string-fret)	One-sided power law
Air jet in wind instruments	Square root (Bernoulli)
Bowed string (stick-slip)	Signum function

Key challenge: these range from smoothly differentiable to discontinuous.

Numerical Simulation

Time-stepping Methods

1. Start with governing PDEs (continuous space and time)
2. **Spatial discretisation** $\rightarrow$ system of coupled ODEs
3. **Temporal discretisation** $\rightarrow$ discrete update rule

For a linear system, the run-time loop is a state-space recursion:

$$\begin{aligned}\mathbf{h}_{n+1} &= \mathbf{A}\mathbf{h}_n + \mathbf{b} f_n \\ \mathbf{y}_n &= \mathbf{C}\mathbf{h}_n\end{aligned}$$

The matrix $\mathbf{A}$ is extremely **sparse** (physics is local).

Stability: Energy Conservation

The primary challenge: ensuring **numerical stability**, especially for nonlinear systems.

Passivity principle: a physical system does not generate energy.

$$\frac{dE}{dt} \leq P_{\text{in}}$$

Energy-conserving schemes: enforce conservation of a discrete energy quantity to machine precision.

⇒ Guaranteed bounded solutions for arbitrary simulation lengths.

Next Week

Session 2: Modal and Finite-Difference Methods

- Eigenvalue problems for vibrating systems
- Modal synthesis of strings, membranes, and plates
- FDTD schemes
- Stability and dispersion analysis
- Boundary conditions
- Both approaches as state-space ODEs

Homework

Reading

- Bilbao, *Numerical Sound Synthesis*, Ch. 2 (time series and finite difference operators) and Ch. 3 (the simple harmonic oscillator).

Optional

- Implement the SHO using the **explicit scheme based on the centred second difference operator δ_{tt}** (Ch. 3). Vary the step size, observe when stability breaks, and compare to the analytical solution.